Mechanical Metrics (MechMet)

Overview

- **MechMet**: **MatLab** code for computing mechanical metrics of Virtual Polycrystals
- Utilizes Virtual Polycrystals created by **Neper**
- Solves field equations for elasticity using the finite element method
- From computed displacement fields, computes several mechanical metrics
- Generates a formatted output file for visualization with **Paraview** or **VisIt**

Mechanical metrics:

- Strain and stress tensors
- Embedded (apparent) directional stiffness
- Isolated (single-crystal) directional stiffness
- Normalized Schmid Factor
- Directional strength-to-stiffness ratio

Mechanical Metrics (MechMet)

MechMet origins and intent:

- Draws on codes/scripts/functions developed at Cornell over the years in:

  MAE courses, **ODFPF** package, **FEpX** pre- and post-processing scripts
- Intended as a relatively 'quick-and-easy' tool for investigating mechanical behaviors of polycrystals, especially in conjunction with *in-situ* loading diffraction experiments
- Not intended for more complex elastoplastic simulations (such as possible with **FEpX**).

MechMet code and interfaces:

- Code runs on laptops (mesh size limited by computer memory)
- 10-node tetrahedral elements w/15 point quadrature rule (same as **FEpX**).
- Code utilizes **MatLab** in-line solvers (direct or iterative; direct recommended)

MechMet itself is a stand-alone **MatLab** FE package that works with two other resources:

1. *Input from:* Virtual polycrystal instantiation tools for microstructure and mesh (**Neper)**
2. *Output to:* Visualization/interpretation tools **(Paraview** or **VisIt)**

Mechanical Metrics (MechMet)

Running MechMet:

- Generate/obtain **Neper** .msh file for the virtual polycrystal (use version 4).
- The single crystal properties are specified in *filename*.matl files. Line 1 is the crystal type (3 for cubic-fcc; 4 for cubic-bcc; 6 for hex); Line 2 lists the elastic moduli; Line 3 lists the slip system strengths. The **Neper** files and the .matl file need not have the same basename. **MechMet** handles multi-phase polycrystals as defined by **Neper.**
- The boundary conditions are specified in *filename*.bcs files. Boundary conditions are specified for extension and/or rotation (for bending loads) surface-by-surface.
- MechMet is executed by running MechMet and providing the input filenames and loading information requested.
- MechMet exports .vtk and .h5 output files for use with either **Paraview** of **VisIt** for visualization

Mechanical Metrics (MechMet)

Mechanical response:

- Linear elasticity w/anisotropic Hooke's Law (for cubic and hexagonal symmetries)
- Loading modes: uniaxial extension in x, y or z and biaxial extension in x-y, y-z, or z-x planes. In addition, rotations may be imposed to simulate bending.
- Solves for displacement field, then computes strain (from grad **u**)
- Computes stress from strain field, together with Hooke's law.

Mechanical metrics:

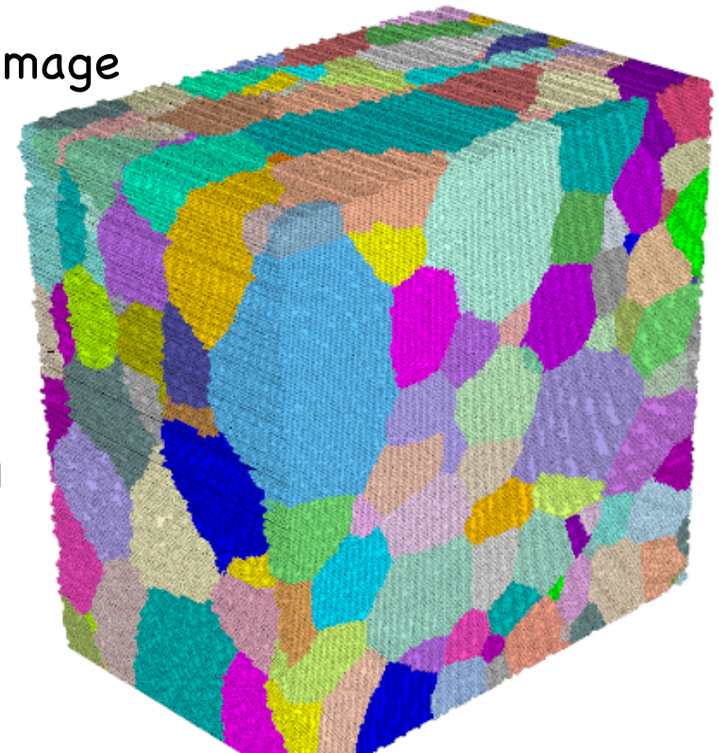
- Single crystal directional stiffness: reciprocal (diagonal of compliance matrix).
- Embedded (apparent) stiffness: ratio of average stress to local (fem) strain.
- Schmid Factor: ratio of resolved shear stress to applied stress (assumes nominal stress).
- Strength-to-Stiffness: critical resolved shear stress / embedded stiffness (Poshadel&Dawson (MMTA,2019))

Mechanical Metrics (MechMet)

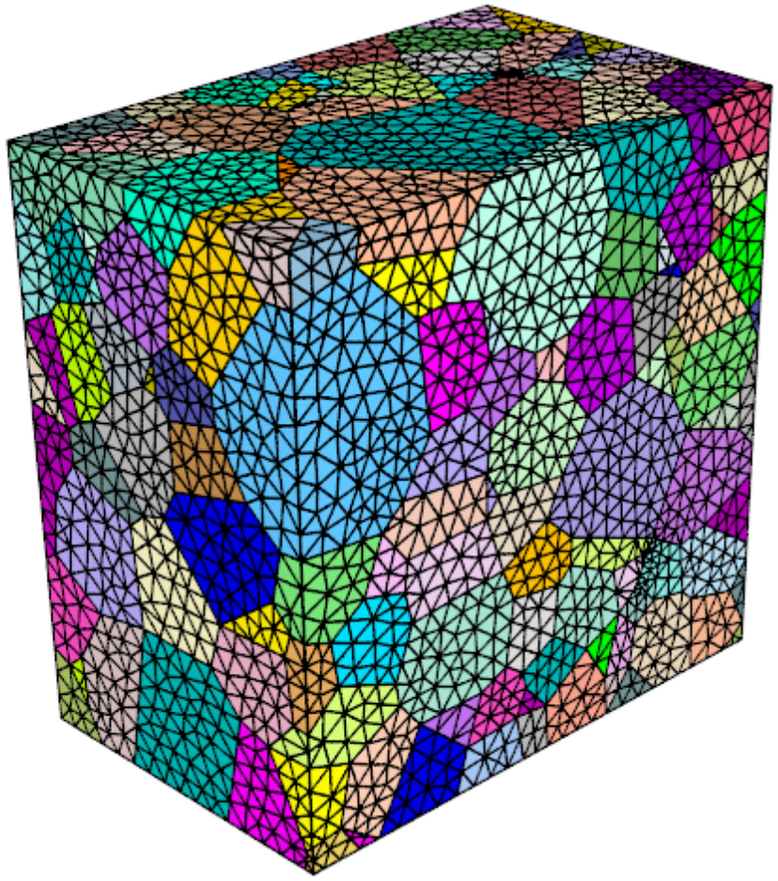
Ti7Al Example:

- HEDM sample with 440 grains (100 pixel threshold)
- Sample data in form of **Neper** .tesr file (D. Pagan)
- S/X properties taken from Ti6Al4V parameters
- Simulations of x, z and x+z direction extension of 0.1% strain
- FE mesh: 81,642 elements and 116,475 nodes

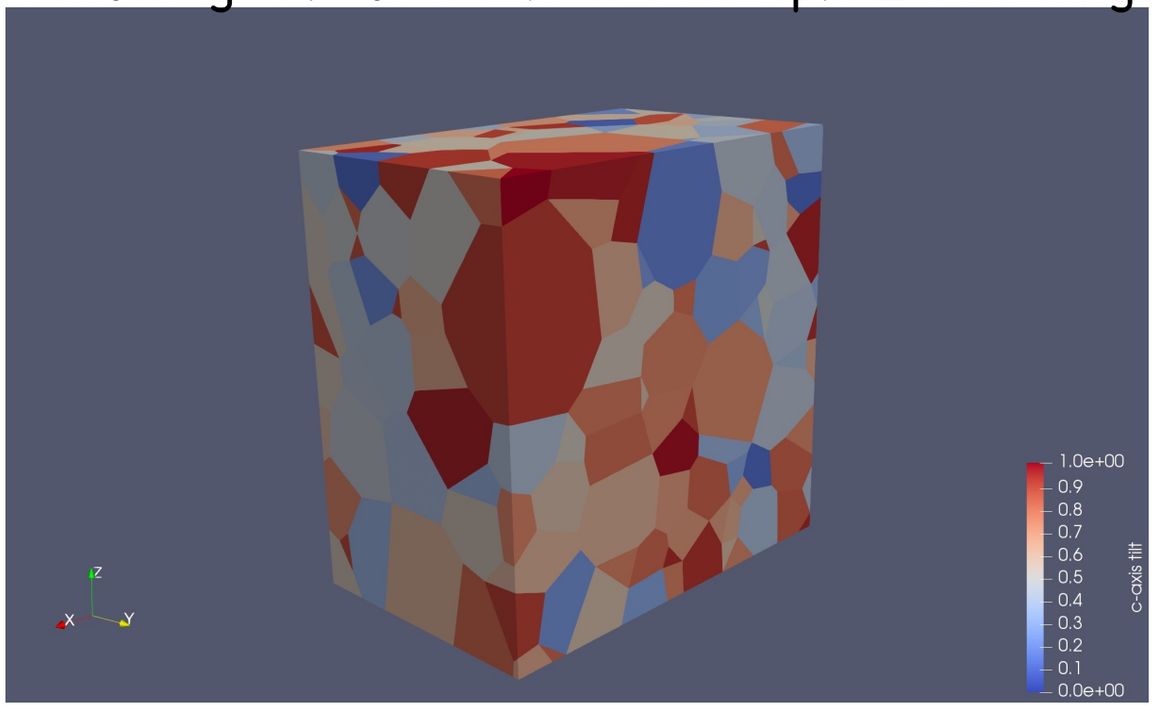
Tesr file image



Tessellation showing grains and mesh



Tessellation showing 'tilt' of c-axis with sample Z in each grain

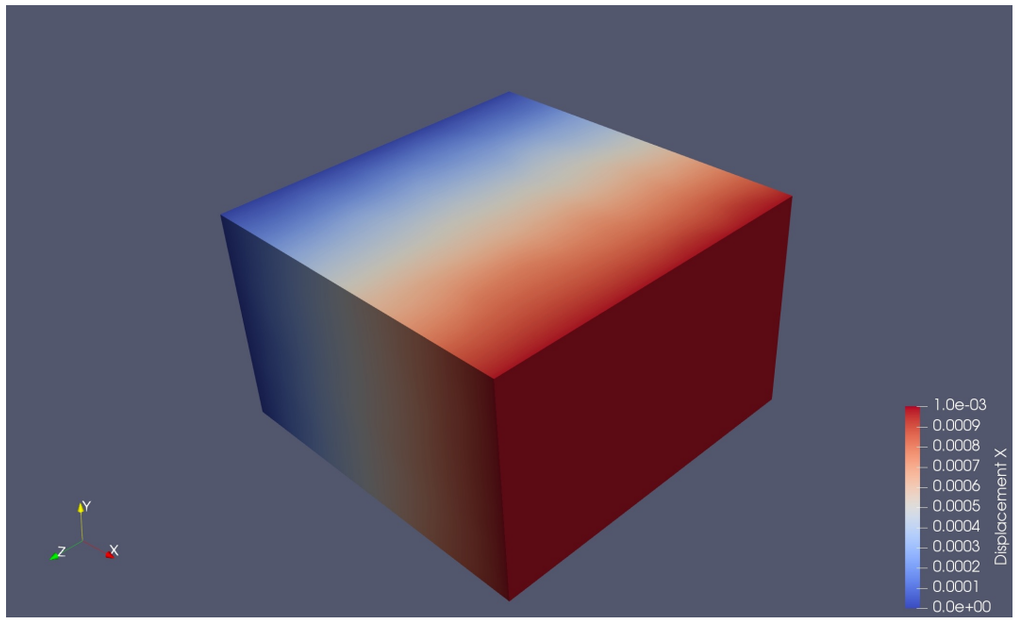


Computed as the mag of x and y components of Rodrigues vectors

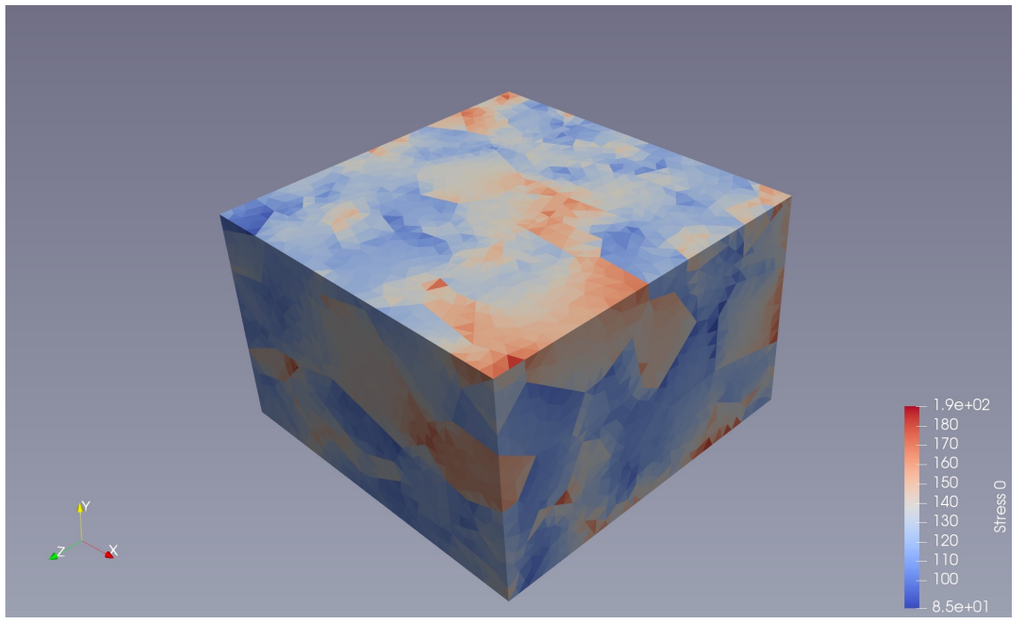
Mechanical Metrics (MechMet)

MechMet solution: Displacement, strain and stress fields

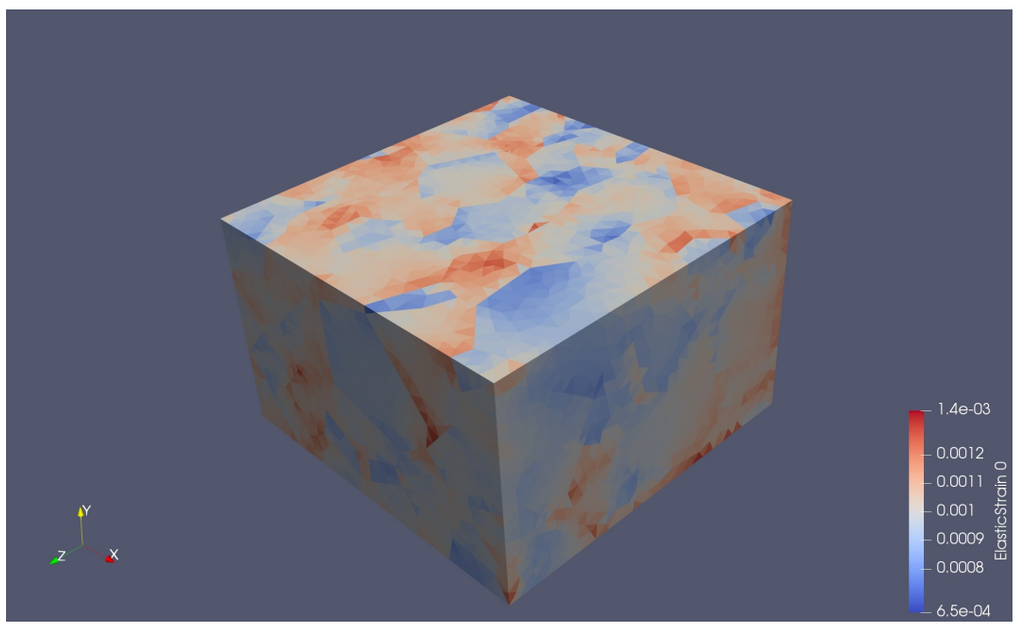
x displacement field



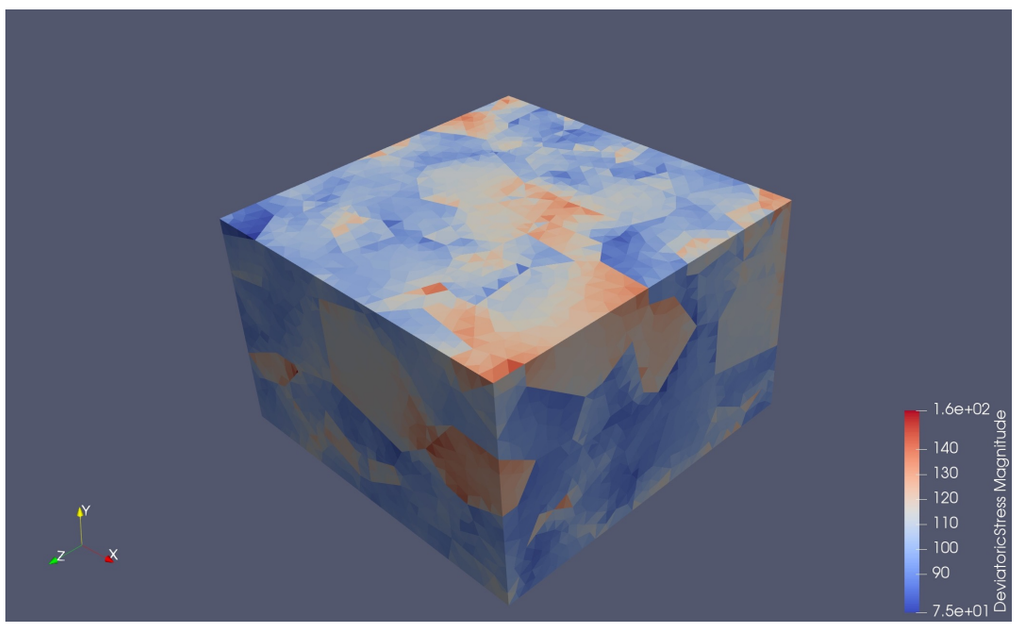
xx component of stress



xx component of strain



vonMises Stress (mag of deviatoric stress)



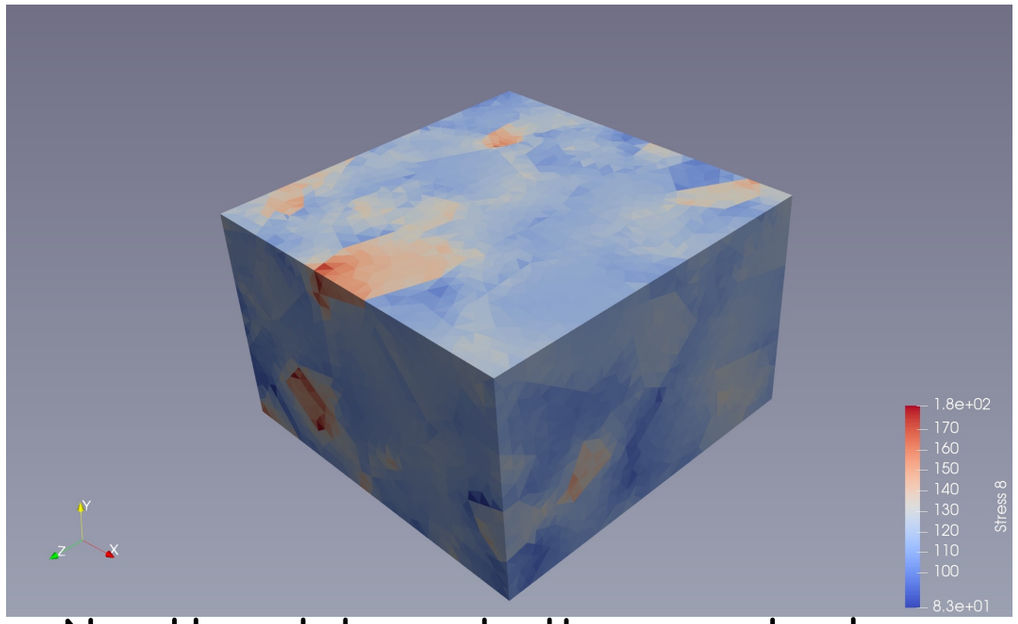
Mechanical Metrics (MechMet)

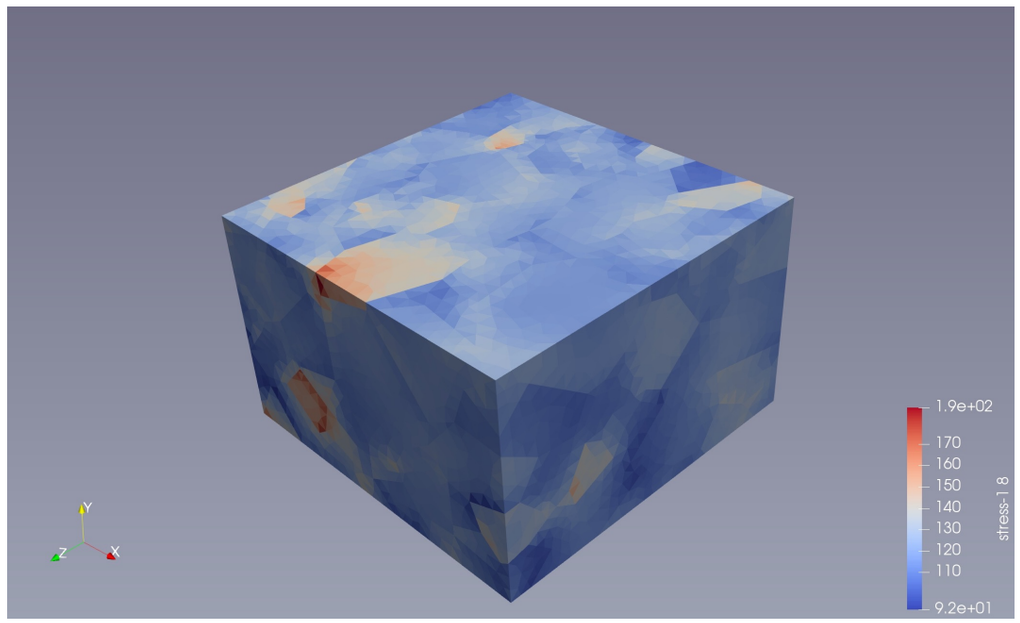
MechMet solution: Comparison to FEpX for Z-extension

MechMet

zz component of stress

FEpX





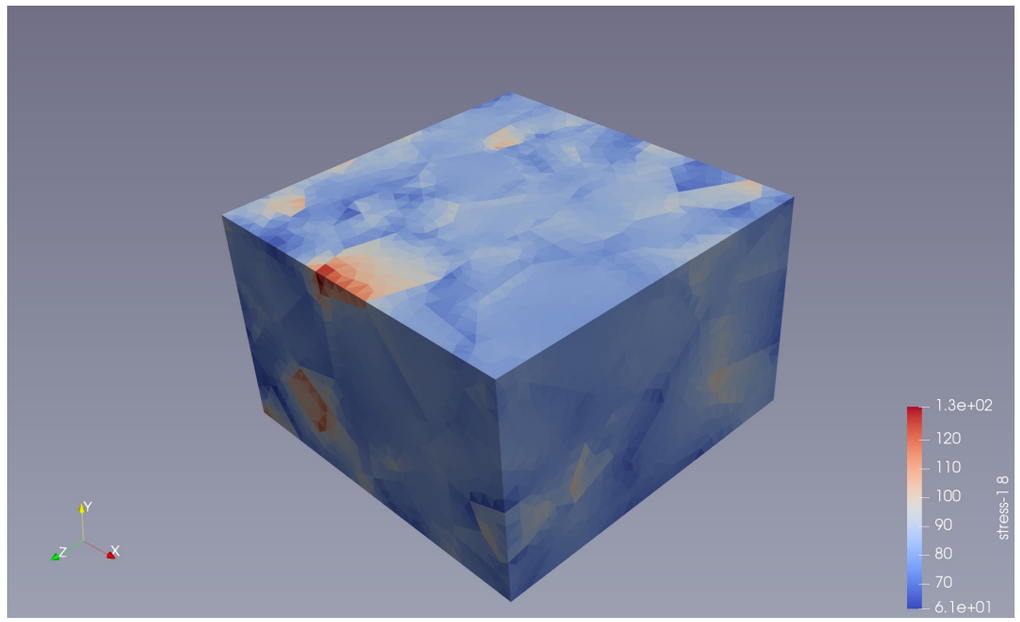
traction-free (non-planar)

- No attempt to apply the same load
- Solution is linear; re-scale as needed.

Two types of BC's applied in FEpX:

- traction-free lateral sides
- planar surfaces w/zero net traction

Actual loading of sample in experiments probably lies between these two idealizations.



planar w/zero net traction

Mechanical Metrics (MechMet)

Stiffness(only)-Based Metrics — Background

- Embedded (apparent) stiffness = ratio of average stress to local (fem) strain
- Single crystal directional stiffness = reciprocal (diagonal of compliance matrix)

SX-E

$$\begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \epsilon_{23} \\ \epsilon_{13} \\ \epsilon_{12} \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & & & \\ S_{12} & S_{22} & S_{23} & & & \\ S_{13} & S_{23} & S_{33} & & & \\ & & & S_{44} & & \\ & & & & S_{55} & \\ & & & & & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{Bmatrix}$$

For uniaxial x-tension (σ_{11} only non-zero stress component). the directional (Young's) modulus (which is $E = \sigma_{11}/\epsilon_{11}$) is found from the reciprocal of the diagonal ot the compliance matrix, S_{11}). Here the modulus is taken directly from the single crystal stiffness or compliance matrix.

Embedded E

Computed directional modulus from simulation results. Ratio of the average (nominal) stress to the strain within a crystal. Here we extract the E from computed response. Embedded modulus accounts for neighborhood on mechanical response. Analogous to the 'diffraction modulus', which is the nominal stress divided by the lattice strain (usually for all crystals on the same crystallographic fiber).

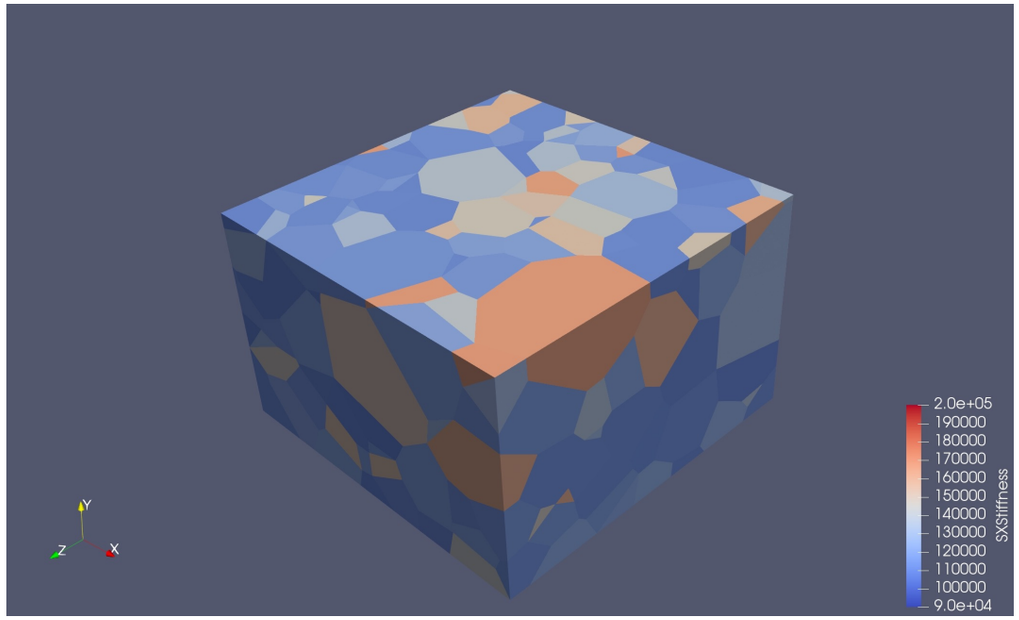
RE2SX

Computed as the ratio: RE2SX = (Embedded E)/(SX-E).
Gives a metric for the response of a crystal within its actual crystallographic neighborhood.

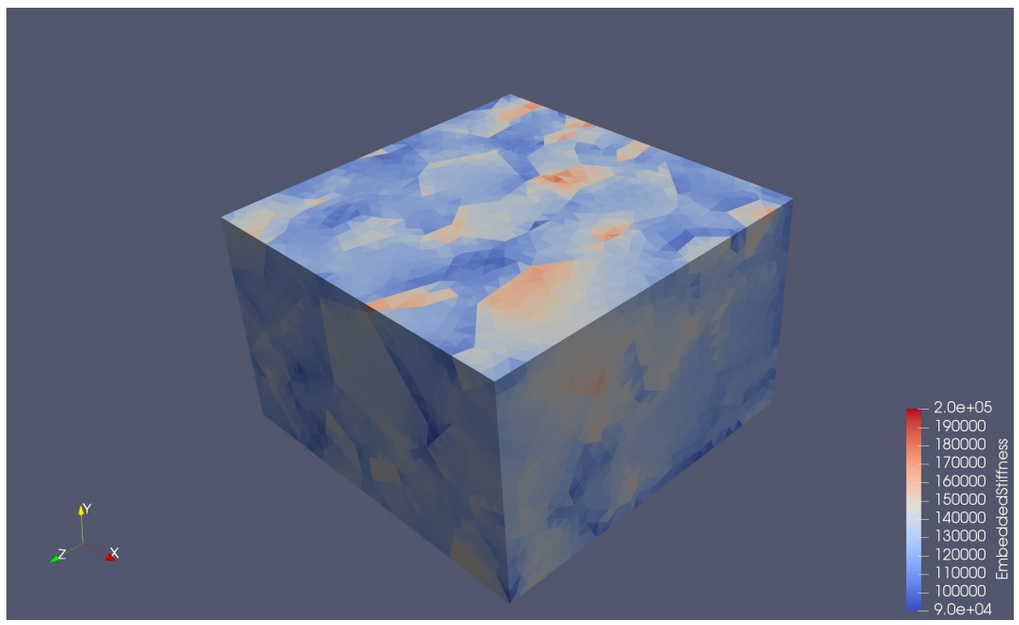
Mechanical Metrics (MechMet)

Comparison of Stiffness Metrics for Two Loading Modes – X Extension

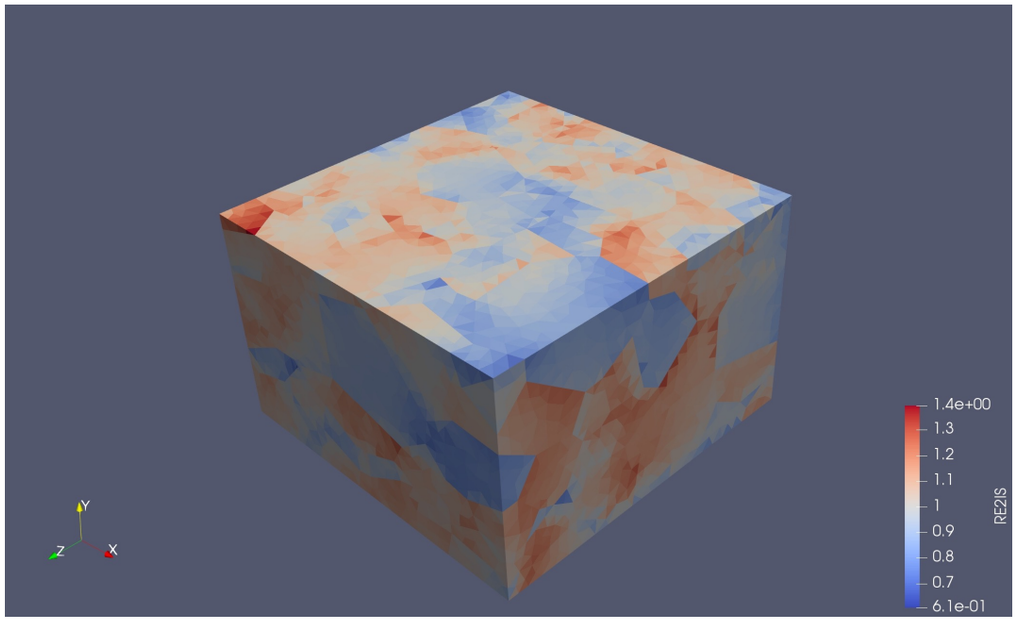
SX-E



Embedded E



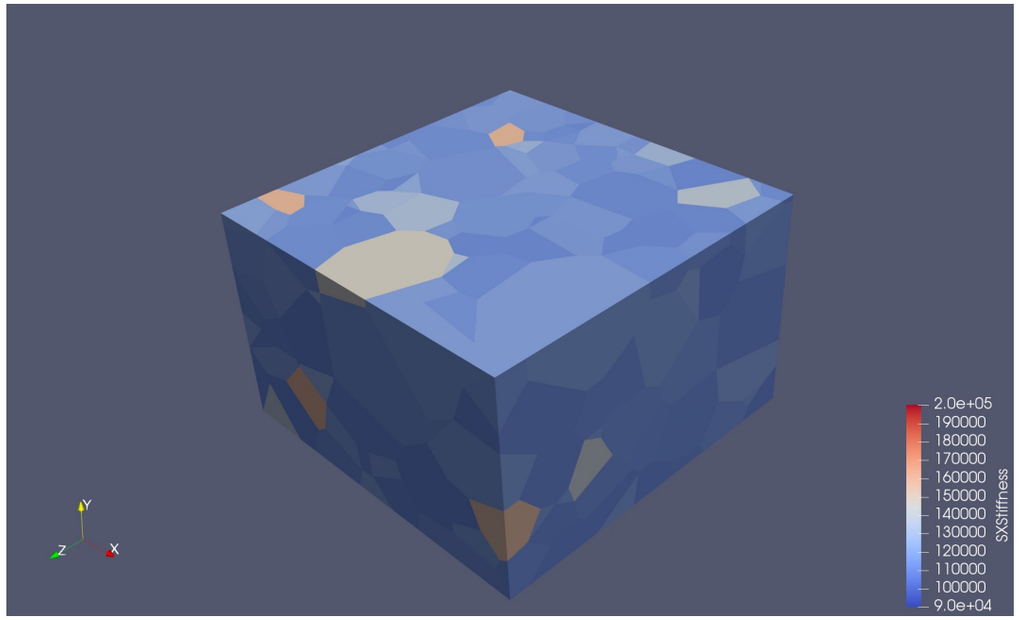
RE2SX
(ratio of embedded to SX stiffness)



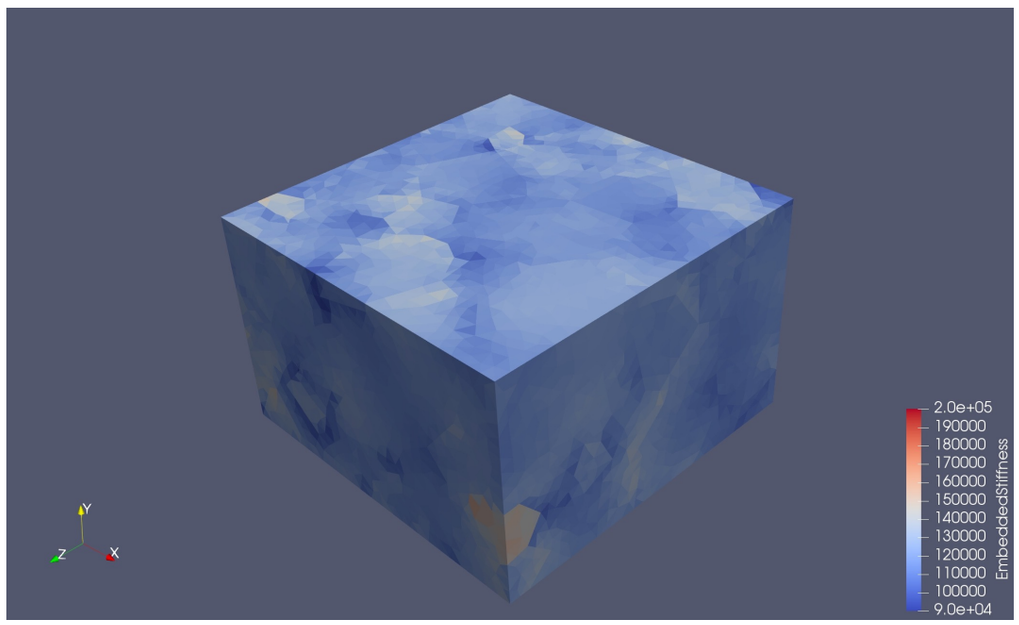
Mechanical Metrics (MechMet)

Comparison of Stiffness Metrics for Two Loading Modes – Z Extension

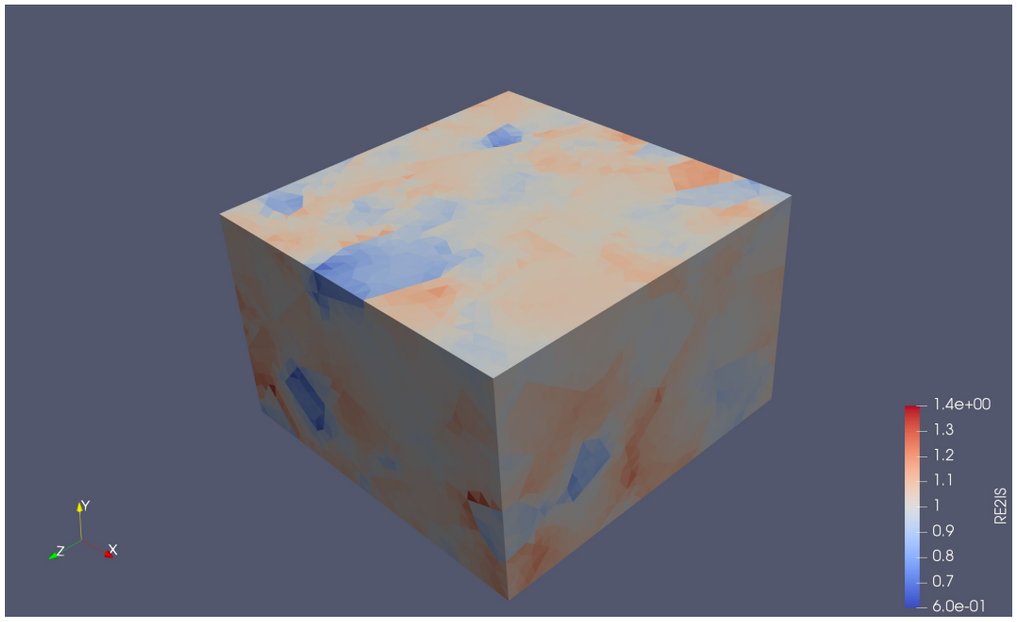
SX-E



Embedded E



RE2SX
(ratio of embedded to SX stiffness)

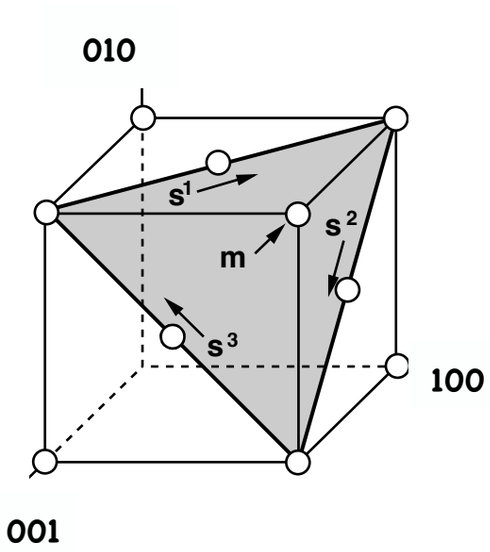


Mechanical Metrics (MechMet)

Strength(+stiffness)-Based Metrics – Background

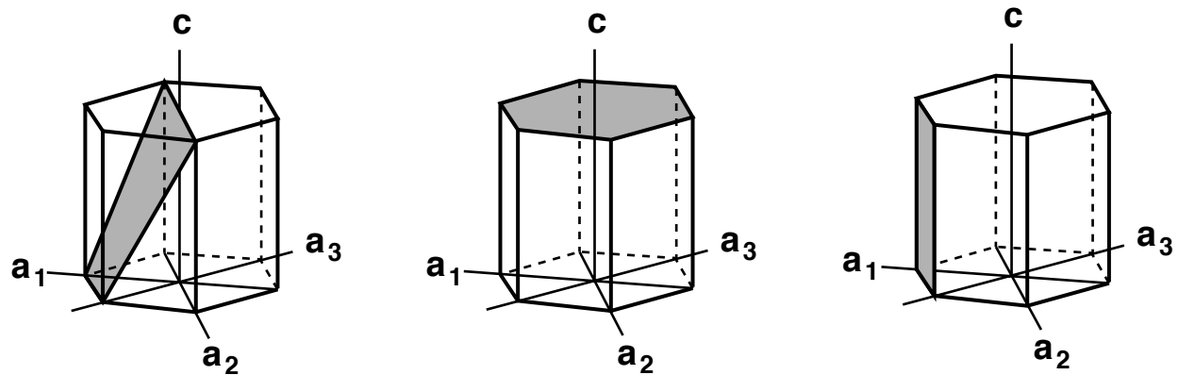
- Slip systems for FCC (octahedral) and HCP (basal, prismatic, and pyramidal) for evaluating Schmid Factor and strength-to-stiffness (Y2E) ratio.
- Schmid Factor assumes nominal stress; Y2E utilizes Poshadel&Dawson (MMTA,2019) approach for extracting local stiffness from FE data.

FCC slip systems



1 family of slip systems: octahedral

HCP slip systems



Pyramidal
m = {10$\bar{1}$1}
s = <11$\bar{2}$3>

Basal
m = {0001}
s = <11$\bar{2}$0>

Prismatic
m = {10$\bar{1}$0}
s = <11$\bar{2}$0>

3 families of slip systems: basal, prismatic and pyramidal

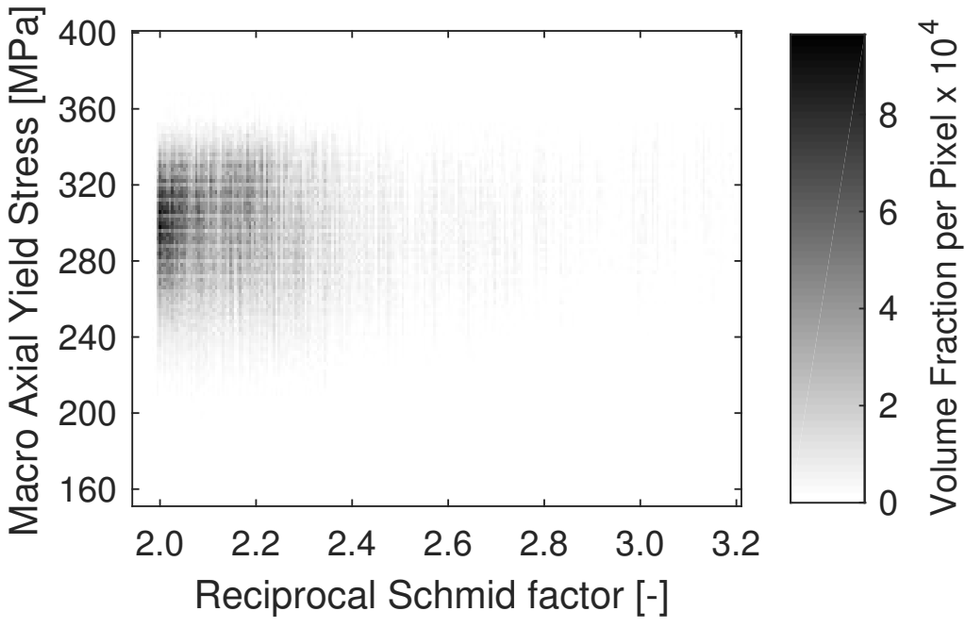
Directional Strength to Stiffness Ratio = Nominal Strain*(Slip System Strength/Actual Resolved Shear Stress)

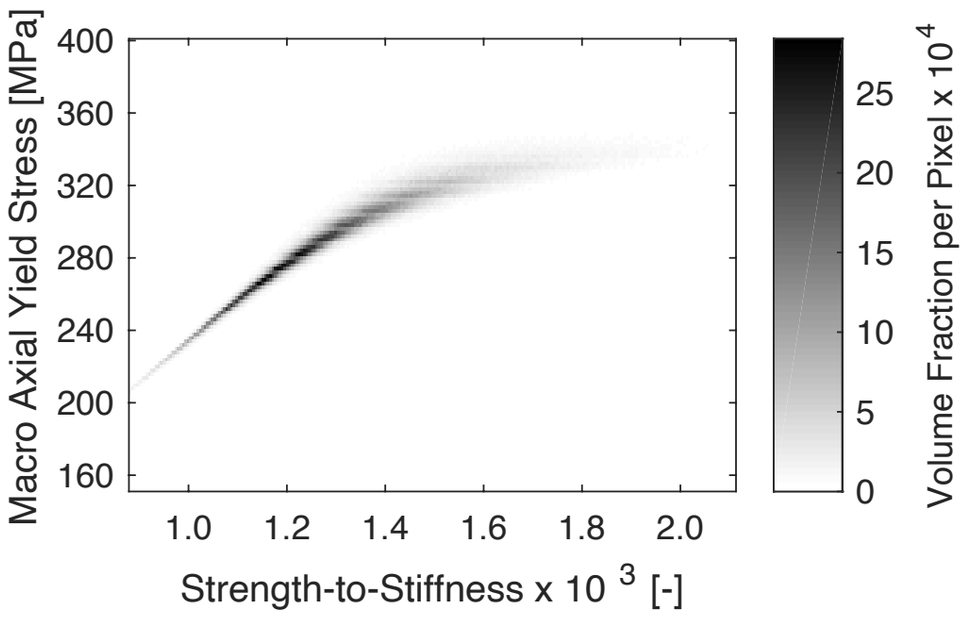
Normalized Schmid Factor = Resolved Shear Stress for Unit Stress / Relative Slip System Strength

= Schmid Factor / Relative Slip System Strength

Mechanical Metrics (MechMet)

Strength(+stiffness)-Based Metrics – Sample Statistics for FCC samples





Example: stainless steel (FCC), having relatively high elastic anisotropy and octahedral slip systems. Response simulated with 5000 grains discretized with 0.5M elements

First point: the higher the Schmid Factor, the more likely a slip system is to be the first to yield in any given crystal.
The reciprocal Schmid factor then gives a relative strength - the higher its value, the less likely that slip system is the first to yield.

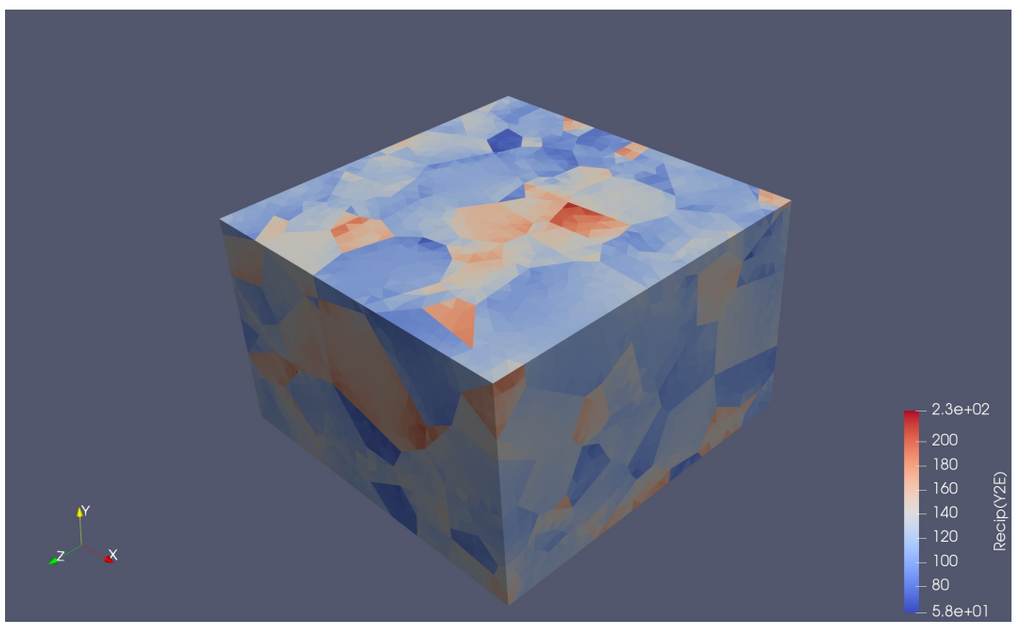
Second point: the strength-to-stiffness metric is a much better predictor of initial yielding than the reciprocal Schmid factor – mainly because of the scatter in the distribution associated with the Schmid factor.

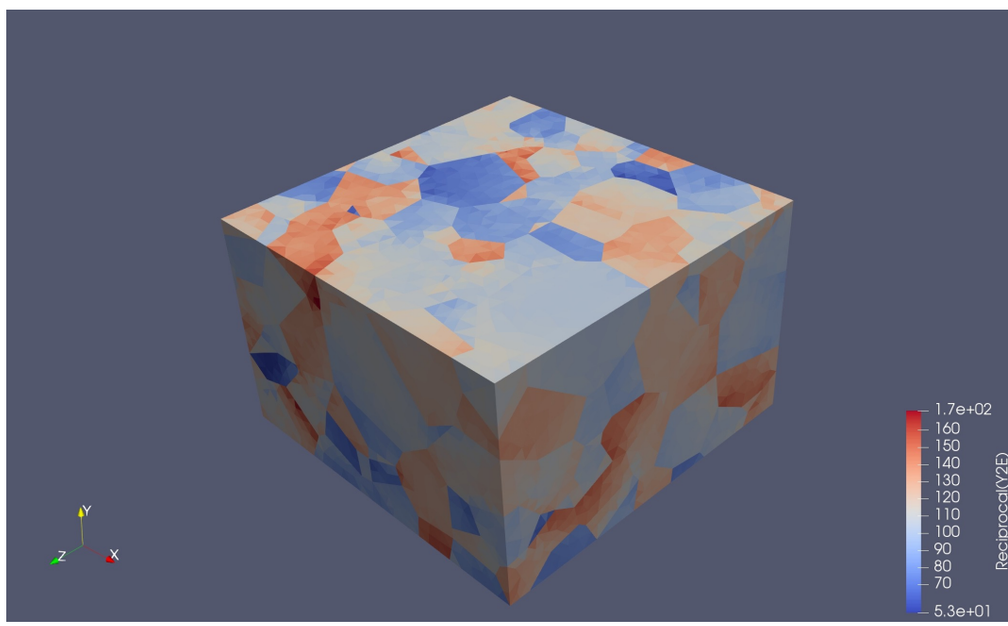
Mechanical Metrics (MechMet)

x-direction extension

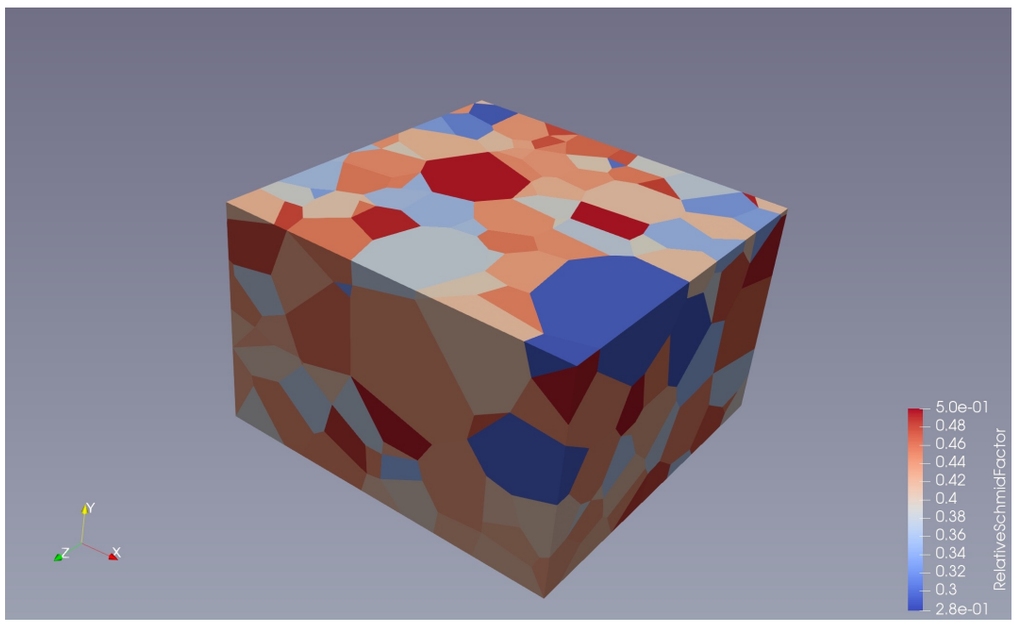
z-direction extension

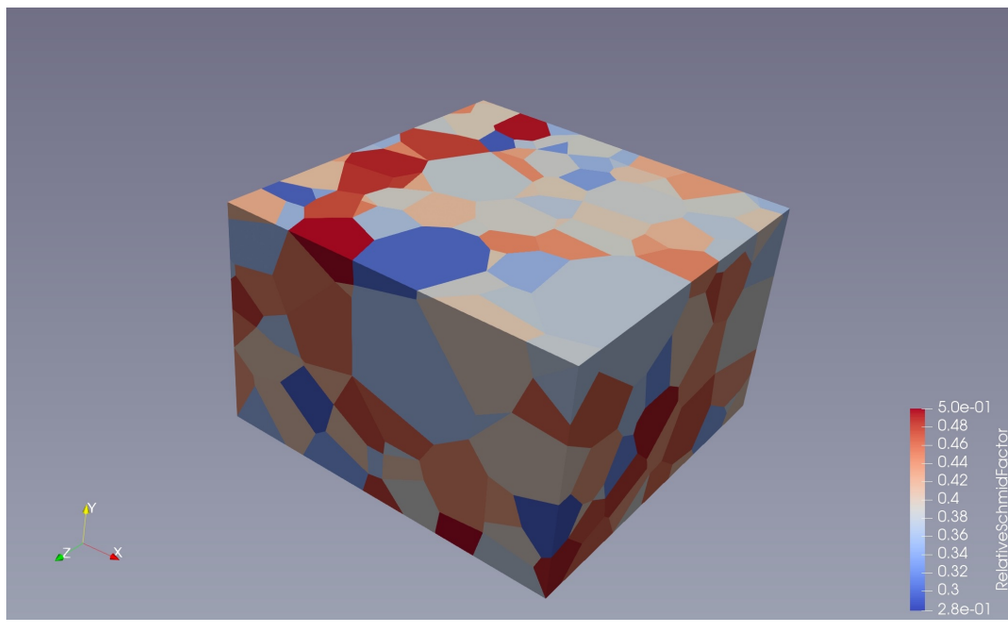
1/Y2E





Normalized Schmid Factor



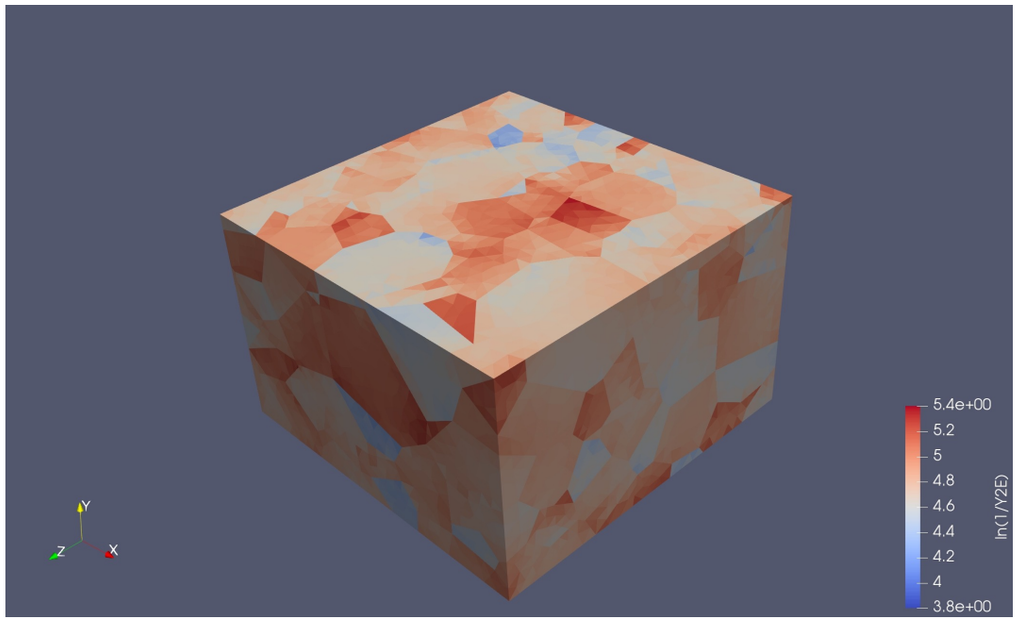


Note: Using 1/Y2E and Schmid Factor means high value correspond to weaker grains (first to yield)

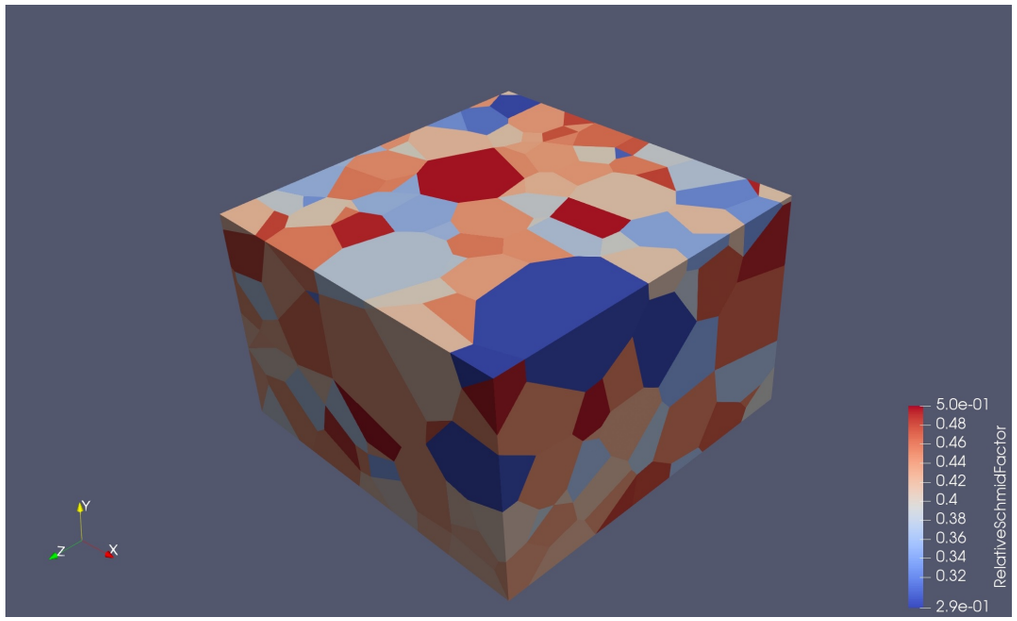
Mechanical Metrics (MechMet)

Comparison of Yielding Metrics for Two Loading Modes – X Extension

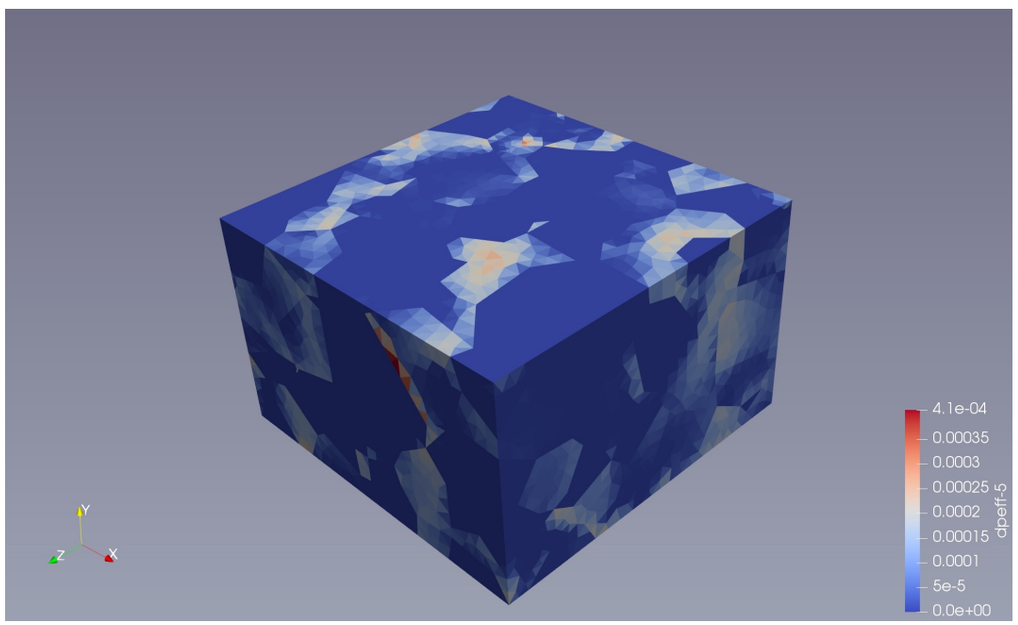
ln(1/Y2E)



Normalized Schmid Factor



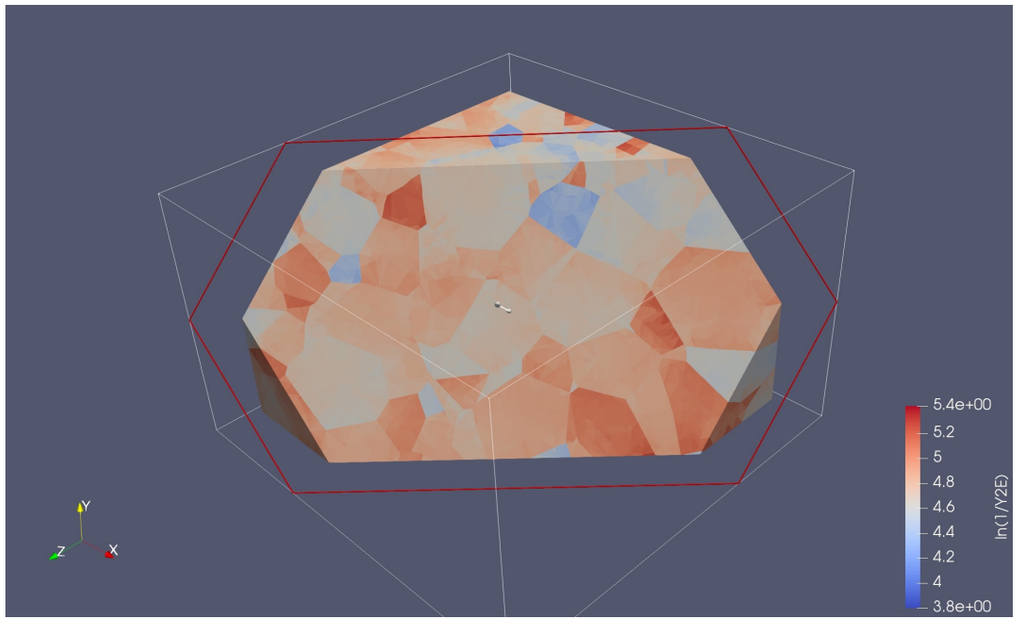
Effective Plastic Strain Rate



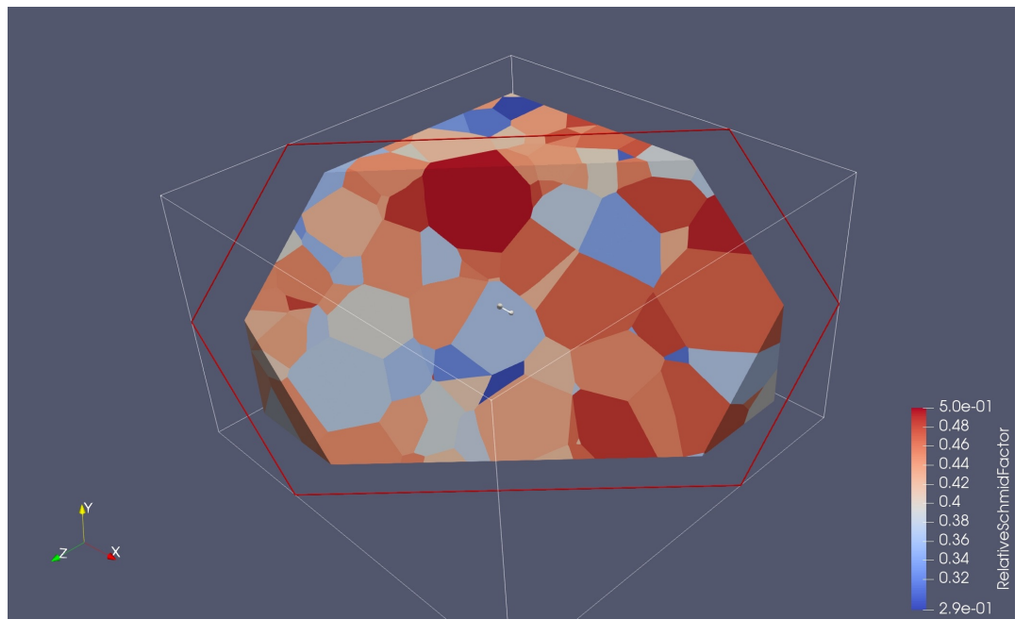
Mechanical Metrics (MechMet)

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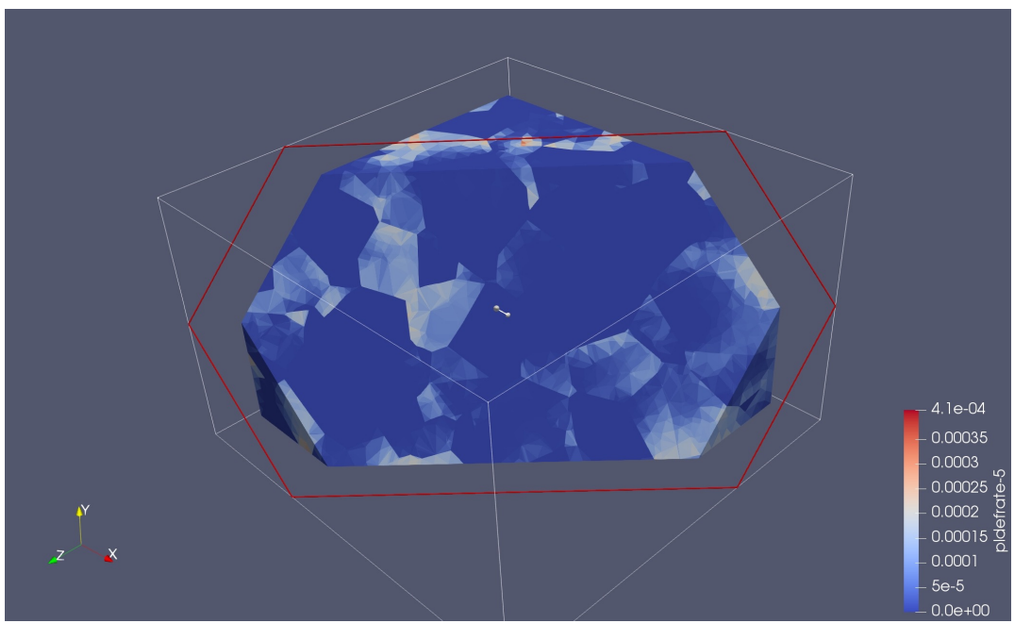
ln(1/Y2E)



Normalized Schmid Factor



Effective Plastic Strain Rate

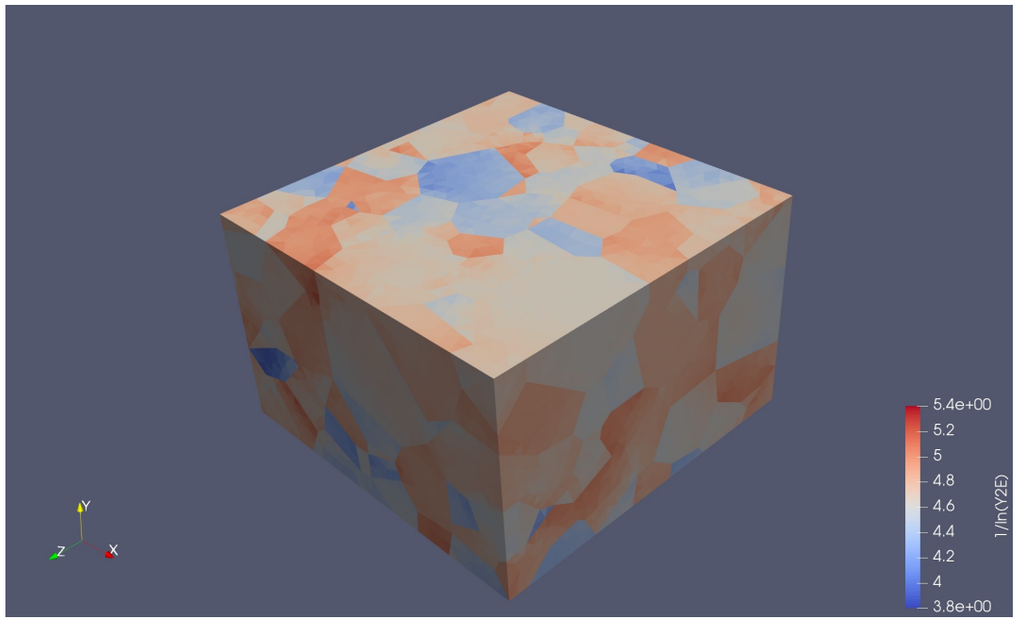


Clipped using plane with (1,1,1) normal passing through the sample centroid

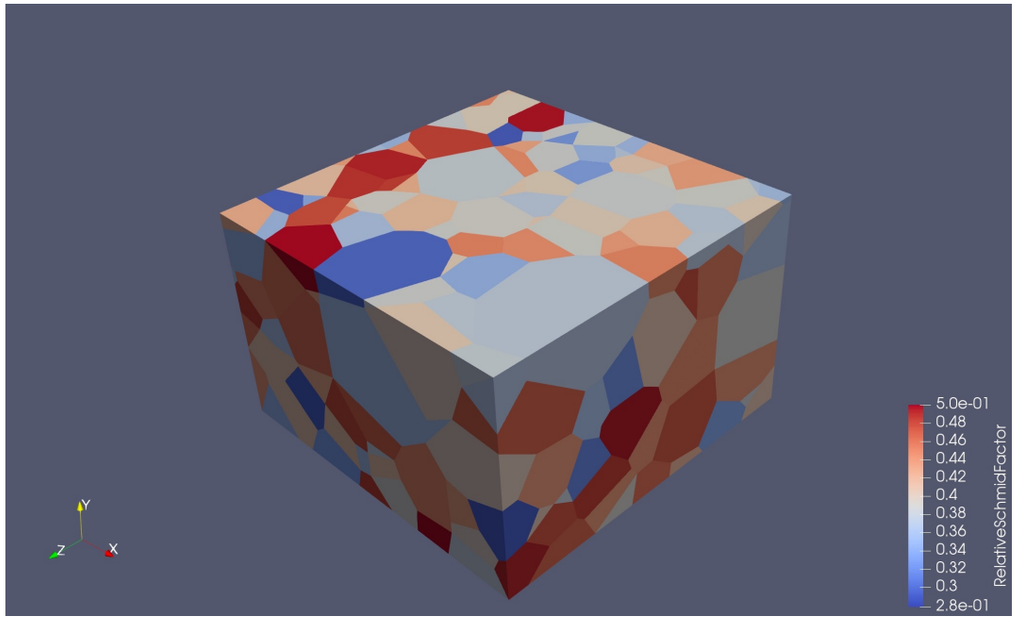
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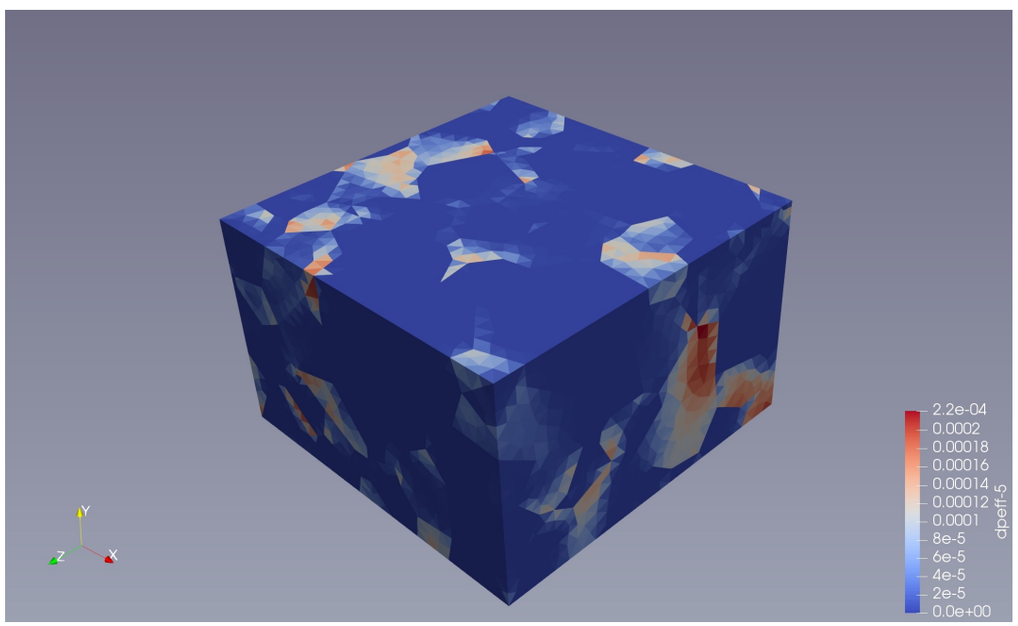
ln(1/Y2E)



Normalized Schmid Factor



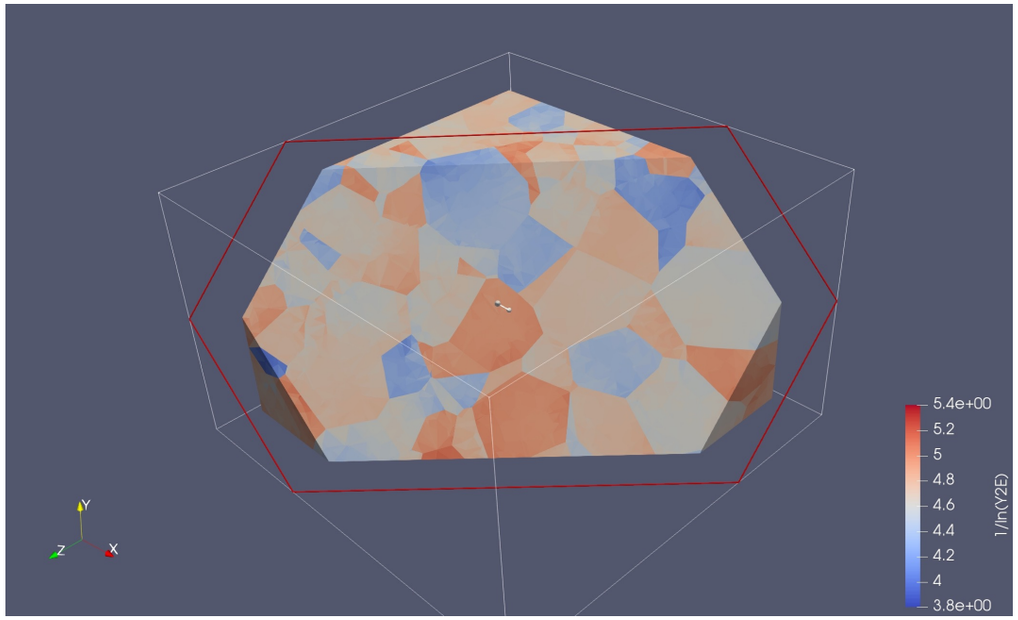
Effective Plastic Strain Rate



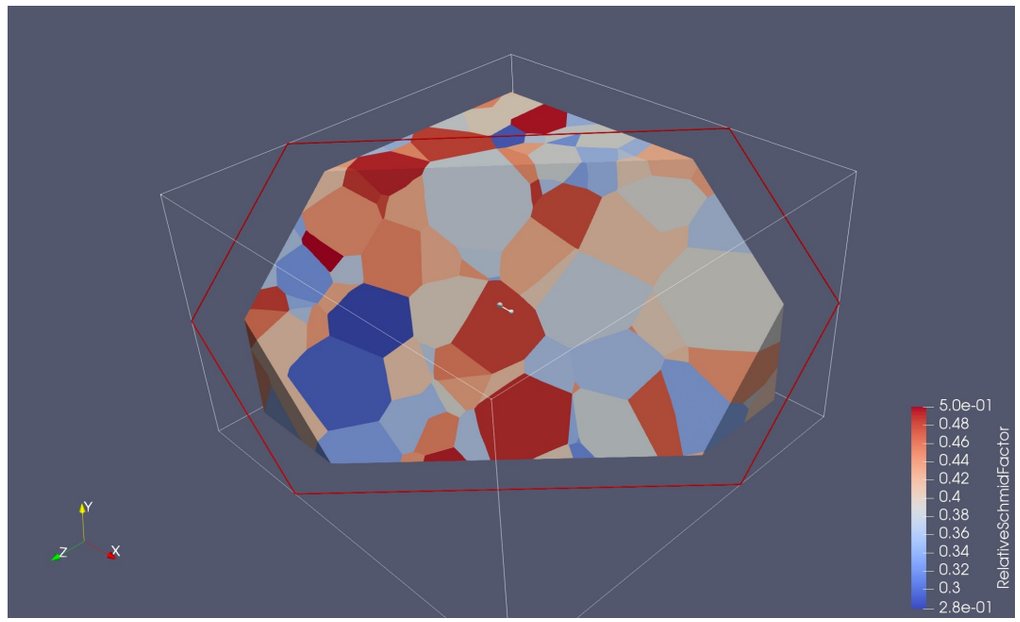
Mechanical Metrics (MechMet)

Comparison of Yielding Metrics for Two Loading Modes – Z Extension

ln(1/Y2E)

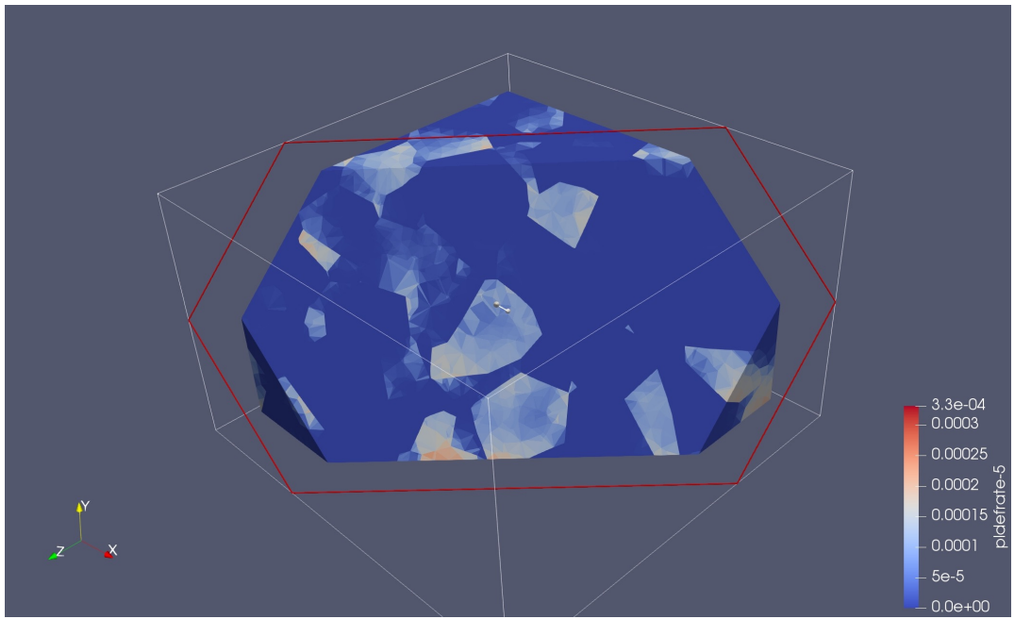


Normalized Schmid Factor



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Effective Plastic Strain Rate



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